

Populations/Samples and Descriptive Statistics

SDS 320E

Introduction to Statistics

Intro to Statistics

- **Statistics** - *the art of collecting, organizing, summarizing, and describing data as well as drawing inferences from the data.* - Dr. Amy Maddox
 - Statistics quantifies our uncertainty about our data i.e Does our results from our sample generalize to the entire population?

Example

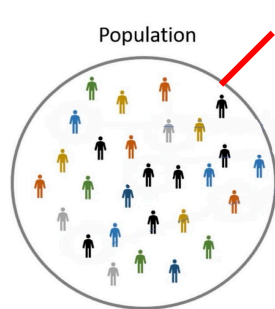
Suppose the FDA requires more than 50% of possible patients to see improvement for a proposed drug to be approved. A pharmaceutical company in Austin, Tx has developed a new Alzheimer's drug which is supposed to improve mental cognition. The company just finished its clinical trial and found 50.5% of those in the sample had improved mental cognition.

1. What are some reasons the drug should be approved?
2. What are some reasons the drug should NOT be approved?

Research Question

! Goal of Statistics

Generally, the goal of statistics is to determine some characteristic about the population of interest.



- This goal is often summarized in a **research question**
- A good research question should:
 - References the population of interest
 - Not be too broad or too narrow

Research Question Cont.

Example

UT wants to launch a new on-campus service and needs to determine if students are interested. UT proposes three research questions given below:

1. How many current UT students are interested in using a new on-campus service?
2. Is the availability of evening hours related to student interest in using a new on-campus service?
3. What impacts student interest in campus services?

Which research question is too broad, which is too narrow, and which is just right?

- 1.
- 2.
- 3.

Population of Interest and Census

- We can only answer a research question with 100% accuracy if a census is taken
- **Census** - *a survey of the entire population of interest*
 - However, what is the issue with a census?
 - Census are very rare and in most cases unavailable

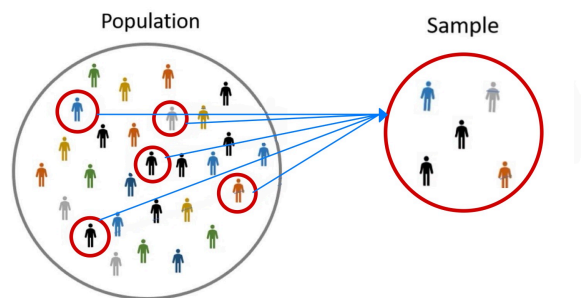
Example

UT wants to determine whether students would be interested in using a new on-campus service that is launching. What is the population of interest?

All current students or all possible students?

When We Don't Have A Census

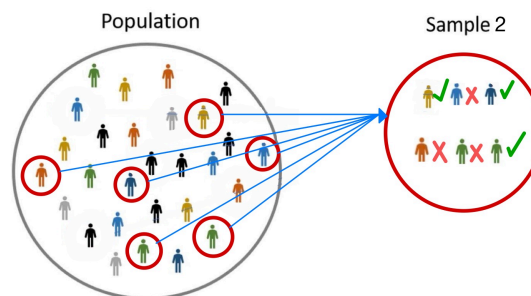
- Since we can not take a survey of the entire population, generally we take a _____
- **Sample** - *a subset of the population of interest*



Sample Cont.

Example

Lets return to our previous example. Suppose a majority of the sample says they would be interested in using a new on-campus service that is launching. Does this guarantee the majority of the population will be interested in using the new on-campus service that is launching?



➤ If we took another sample, are we going to get the exact same results?

Why Statistics is Hard

➤ In statistics we are trying to solve a really hard and nearly impossible problem

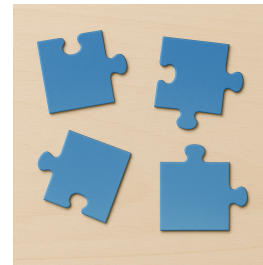
➤ **Goal:** Infer something about the population given just one sample

💡 Example

Suppose two students are each given a few puzzle pieces and are asked to guess what the puzzle is about.



Student 1 Pieces



Student 2 Pieces

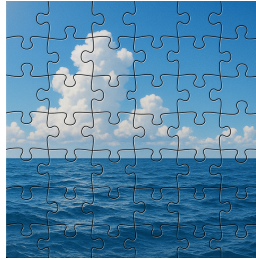
➤ Student 1 claims the puzzle makes an ocean

➤ Student 2 claims the puzzle makes the sky

Which student is correct?

Why Statistics is Hard Cont.

- Again, it is impossible to know without taking a *census*



Full Puzzle

- Both students were partially correct and made reasonable inference based off their puzzle pieces (data)
 - Now imagine the total number of puzzle pieces is in the millions, billions, or even trillions. This becomes a very hard problem!
 - This is what we are dealing with in statistics. Using a relatively small sample to make inference about an entire population
-

Data and Statistics

What is Data?

- To understand data manipulation, visualization, and applied methodology, we must understand what we mean by “data”
- **Data** *is generally any collection of information*
- In statistics, generally, data consist of n observations and p variables (features)

Observations	Variable 1	Variable 2	Variable 3	...	Variable p
Obs 1	23	34	28	...	30
Obs 2	47	23	30	...	35
Obs 3	44	21	32	...	28
Obs 4	54	57	50	...	60
⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮
Obs n	24	28	33		35

- Data is often collected in spreadsheet form via .csv files (Comma Separated Values)

What is Data? Cont.

Example

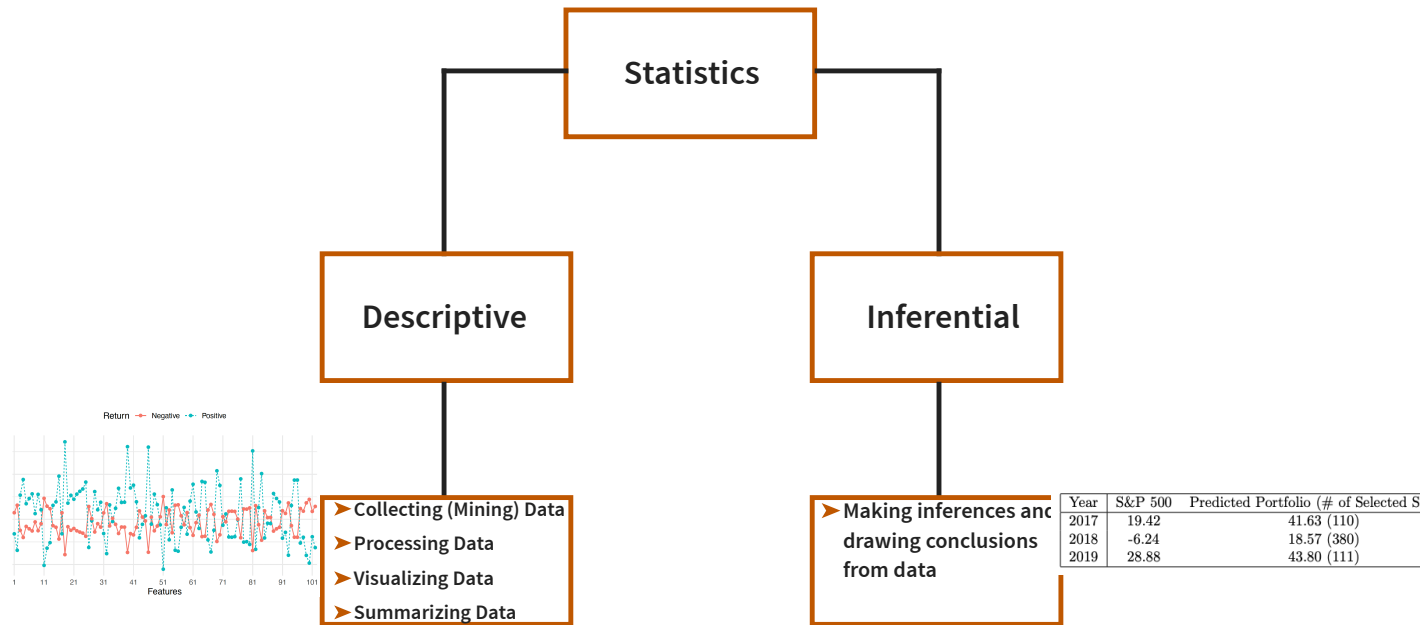
Algorithm stock trading has risen in popularity with AI, however the “buy and hold” strategy often still outperforms algorithms. Thus, is it possible to determining good stocks to “buy and hold” by just a companies 10-k filings? We consider stock return data for a companys’ previous 2017 10-k filing with their 2018 end of year stock return.

ticker	sector	% change	revenue	ebit	gross margin	...
CMCSA	Consumer Cyclical	32.79457	9.45070e+10	1.8653e+10	1.0000	
KMI	Energy	40.58807	1.41440e+10	4.1130e+09	0.4847	
INTC	Technology	30.29551	7.08480e+10	2.3191e+10	0.6173	
MU	Technology	64.21374	3.03910e+10	1.4645e+10	0.5887	
GE	Industrials	44.75784	1.21615e+11	-1.6713e+10	0.2151	
BAC	Financial Services	44.31599	9.12470e+10	3.4584e+10	1.0000	

- 2, 716 stocks or companies
- 103 variables (features) consisting of 10-k filing statement metrics
- The target variable of interest is the % Change (stock return)

Branches of Statistics

- The field of statistics can be generalized into two categories



Types and Sources of Data

The methods and analyses we apply to data and the conclusions we can draw depend on:

- What kind of data we have
 - Is it qualitative or quantitative?
 - On what scale was the data measured?
 - Where or how we obtained our data
 - Is it experimental or observational?
 - Is it population (census) data or just from a sample?
 - This is for both Descriptive & Inferential statistics
 - Different data types will require different graphs and different analyses
-

Types and Sources of Data Cont.

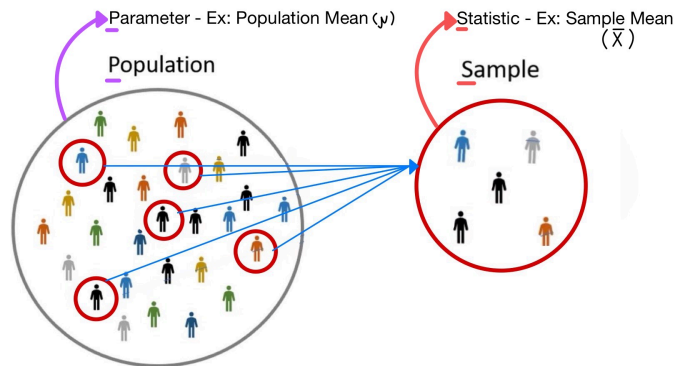
- **Qualitative data** *is data that is categorical in nature* also referred to as just categorical data
 - True/False, Male/Female
 - College (Engineering, Business, Natural Sciences)
 - Education level (High school, Bachelor's, Master's, PhD)

} **nominal data** *has no ordering*
 } **ordinal data** *has meaning in the order*
- **Quantitative data** *is data that is numerical in nature*
 - Currency (\$, €, and ect.)
 - Height, weight, distance, time, and ect.
 - Number of ...
 - Sales made
 - deaths
 - followers gained

} **Continuous data** *is NOT countable*
 } **Discrete data** *is countable*

Measures of Data: Parameter vs Statistic

- When we take a measure of data, we are dealing with either a *parameter* or a *statistic*
- **Parameter:** *a measure of the population*
 - A fixed value (constant) and generally unknown
 - When is a parameter known?
- **Statistic:** *a measure of the sample*
 - A *random variable*, i.e. it will vary from sample to sample



Descriptive Statistics

Measures of Data: Central Location

- Measure of central location describe the “typical” value in a dataset
 - Mean (Arithmetic) `mean()`
 - Median `median()`
 - Mode `DescTools::Mode()`
 - Geometric Mean `DescTools::GeoMean()`
-

Measures of Data: Mean

➤ **Mean** (Arithmetic): *average value of data*

Population Mean

$$\mu = \sum_{i=1}^N \frac{X_i}{N}$$

Sample Mean

$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n}$$

```
1 data <- c(1, 2, 2, 5, 6)
2 mean(data)
```

```
[1] 3.2
```


Measures of Data: Median

➤ **Median:** middle value of data

```
1 data <- c(1, 2, 2, 5, 6)
2 median(data)
```

```
[1] 2
```

➤ If # of observations is even, then the median is average of the two middle values

```
1 data_2 <- c(1, 2, 5, 6) # Even number of observations
2 median(data_2)
```

```
[1] 3.5
```

```
1 (2+5)/2
```

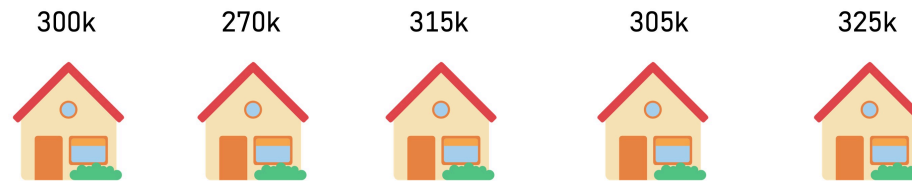
```
[1] 3.5
```

Measures of Data: Mean vs Median

- Mean and Median both describe the “typical” value or center of a dataset
- Why use both? or When is one preferred?

Example

Ex: Suppose as a real estate investor, we are interested in evaluating a newly built neighborhood.



We determine the Mean is \$303,000 and the Median is \$305,000. Which measure should we use to describe the “typical” value of a home in the neighborhood?

Measures of Data: Mean vs Median

What if we changed the value of one of the homes?

Example

Ex: Suppose as a real estate investor, we are interested in evaluating a newly built neighborhood.



We determine the Mean is _____ and the Median is _____. Which measure should we use to describe the “typical” value of a home in the neighborhood?

➤ **Resistant Measure** - A measure that is not affected by outliers

➤ The median is a resistant measure but the mean is not

Measure of Data: Locations

- **Percentile:** The m^{th} percentile, P_m , is the value x such that $m\%$ of the data values are less than x and $(100 - m)\%$ are greater than x .
 - percentile provide measure of specific locations

Question

If you have your IQ tested, would you rather be told you are in the 98th percentile or the 2nd percentile?

```
1 data <- c(1, 2, 2, 4, 5, 6)
2 quantile(data, probs = 0.98)
```

98%
5.9

Measure of Data: Locations Cont.

➤ **Quartiles:** percentiles that divides the data into quarters

➤ $Q_1 = P_{25}$

➤ $Q_2 = P_{50}$

➤ $Q_3 = P_{75}$

```
1 data <- c(1, 2, 2, 4, 5, 6)
2 quantile(data, probs = c(0.25, 0.50, 0.75))
```

```
25% 50% 75%
2.00 3.00 4.75
```

Measures of Data: Variability

► **Variance:** average value the data varies from the mean in units²

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$$

Sample Variance

$$s^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{n - 1}$$

```
1 data <- c(1, 2, 2, 4, 5, 6)
2 var(data)
```

```
[1] 3.866667
```

► **Standard Deviation:** square root of variance i.e. average value the data varies from the mean
in actual units

Population Standard Deviation

$$\sigma = \sqrt{\sigma^2}$$

Sample Standard Deviation

$$s = \sqrt{s^2}$$

```
1 sd(data)
2 ## [1] 1.966384
3 sqrt(var(data)) #i.e. sqrt(3.867) from above
4 ## [1] 1.966384
```

Measures of Data: Variability

- Inner Quartile Range (IQR): variability of the middle 50% of data values

$$IQR = Q_3 - Q_1$$

```
1 data <- c(1, 2, 2, 4, 5, 6)
2 IQR(data)
[1] 2.75
```

```
1 # same as IQR()
2 quantile(data, probs = 0.75) - quantile(data, probs = 0.25)
75%
2.75
```

- Which is measure of variability is a resistant measure?
 - Variance/Standard Deviation
 - IQR

Bivariate Descriptive Statistics

Bivariate Data

- The past measures are for one variable

X	
-0.63	$\mu_X = 0.128$
0.18	$\sigma_X = 0.93047$
-0.84	
1.60	
0.33	

- However, we are often interested in how one variable varies with another
- When we have two variables we say we have **bivariate data**

X	Y	
-0.63	0.21	
0.18	0.18	
-0.84	0.69	
1.60	0.38	
0.33	0.77	Relationship between X and Y = ??

Measures of Data: Variability Between Two Variables

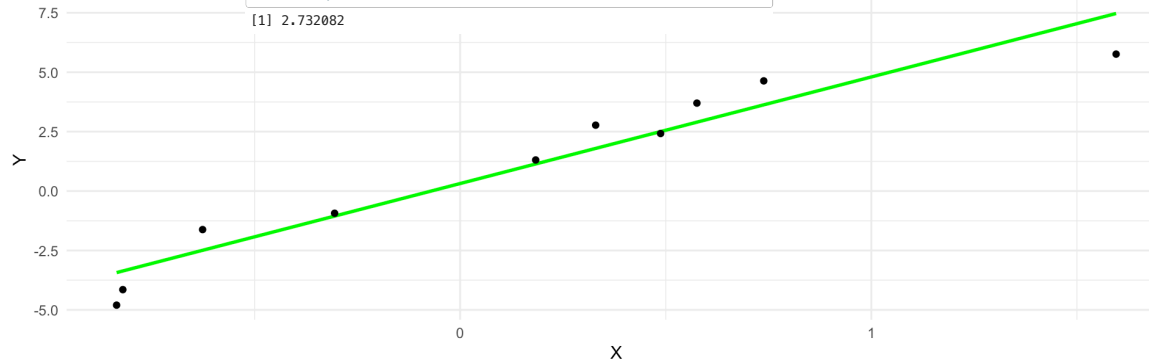
➤ **Covariance:** measure of the linear variability between two variables X and Y

$$\sigma_{xy} = \text{Cov}(X, Y) = \frac{\sum_{i=1}^N (X_i - \mu_x)(Y_i - \mu_y)}{N}$$

➤ + Covariance: when X is above its mean, Y also tends to be above its mean i.e they move in the same direction

► Show code

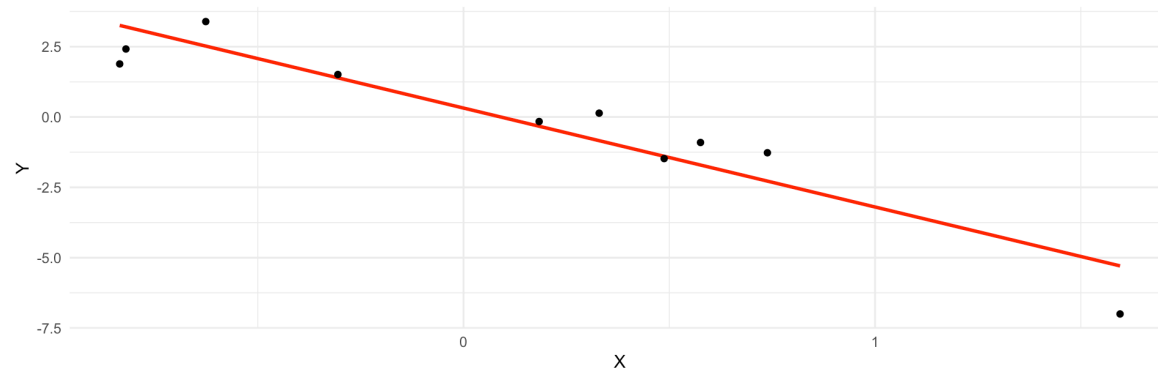
```
1 set.seed(1) # random number generator seed
2 X <- rnorm(10) # random 10 numbers
3 Y <- 5*X + rnorm(10) # Y is 5 times X plus a little error
4 cov(X, Y)
```



Covariance Cont.

- — Covariance: when X is above its mean, Y tends to be below its mean they move in opposite directions

► Show code



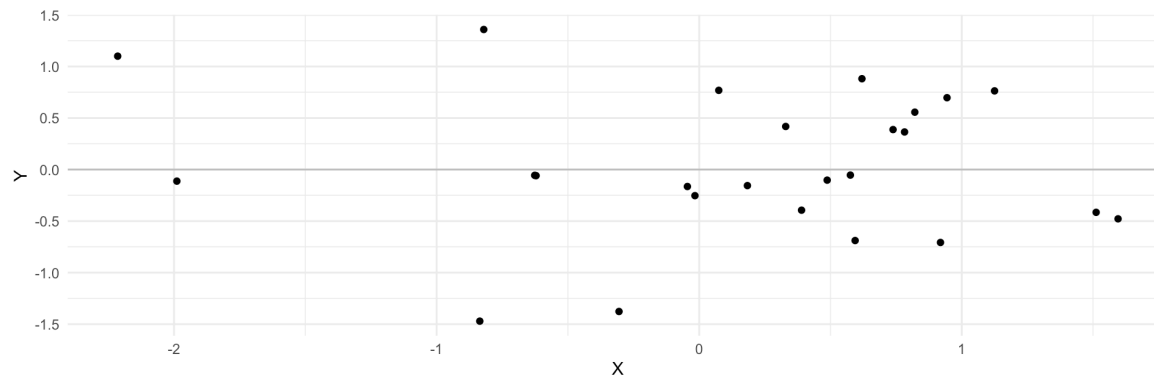
```
1 set.seed(1) # random number generator seed
2 X <- rnorm(10) # random 10 numbers
3 Y <- -3*X + rnorm(10) # Y is - 3 times X plus a little error
4 cov(X, Y) # negative covariance
```

```
[1] -2.142433
```

Covariance Cont.

- Zero Covariance: deviations of X from its mean give no linear information about deviations of Y from its mean.

► Show code



```
1 set.seed(1) # random number generator seed
2 X <- rnorm(25) # random 25 numbers
3 Y <- rnorm(25) # another random 25 numbers
4 cov(X, Y) # close to zero, so very little covariance
```

```
[1] -0.04821342
```

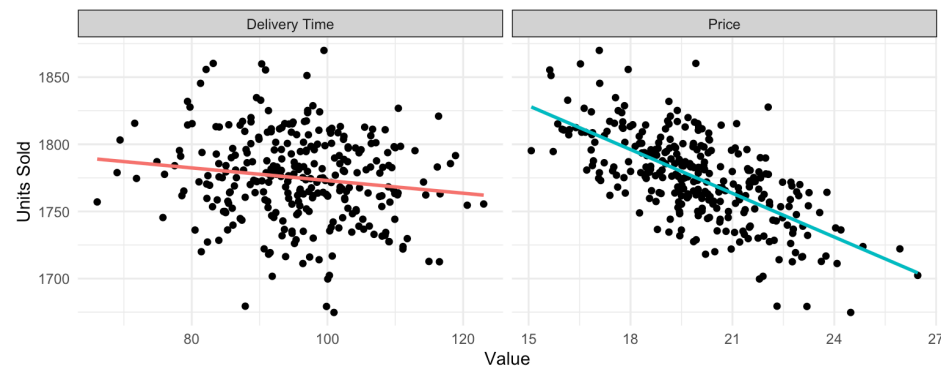
Issue with Covariance

- *Reminder:* Covariance tells us whether two variables are linear related and in which direction
- But what does covariance tell us about the strength of the relationship?

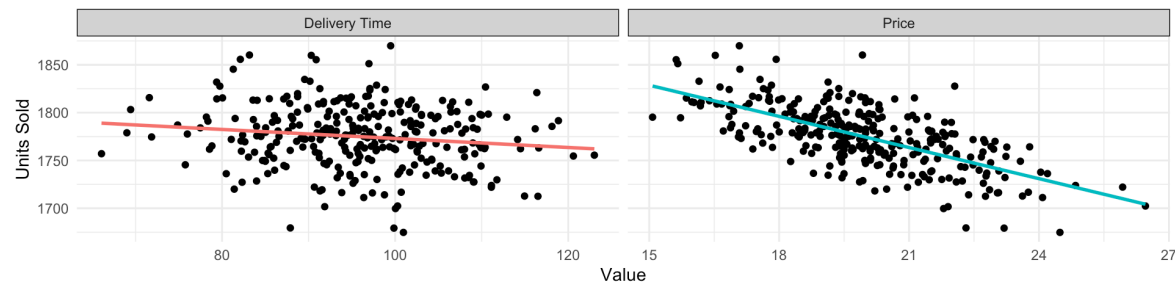
Example

Suppose a company wants to determine whether the delivery time (in hours) or price of a product (in \$'s) has a stronger relationship with units sold. Which variable has a stronger relationship with units sold?

➤ Plot ➤ Data



Issue with Covariance Cont.



Question

Since _____ had a visually stronger relationship with units sold, which variable would we expect to have a larger covariance magnitude?

But...

```
1 cov(price, units_sold)
```

```
1 cov(delivery_time, units_sold)
```

Correlation

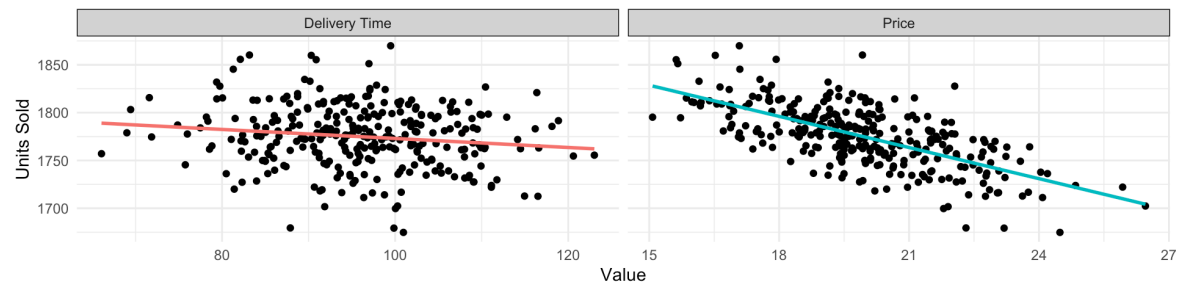
- Covariance tells the direction of the relationship between variables, but we can not compare covariances unless they are in the same units
 - However, we can “standardize” the covariances to make them comparable
- **Correlation:** “Standardized” measure of the linear variability between two variables X and Y

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\sigma_x^2 \times \sigma_y^2}},$$

where $-1 \leq \rho_{XY} \leq 1$. `cor()` in R sample correlation is denoted r

- $-1 \leq \rho < 0$ indicates a **negative** linear relationship
- $0 < \rho \leq 1$ indicates a **positive** linear relationship
- $\rho = 0$ indicates no linear relationship
- ± 1 indicates a perfect negative or positive linear relationship

Correlation Example



Now, from our previous example,...

```
1 cor(price, units_sold)
```

```
[1] -0.6596195
```

```
1 cor(delivery_time, units_sold)
```

```
[1] -0.1429894
```

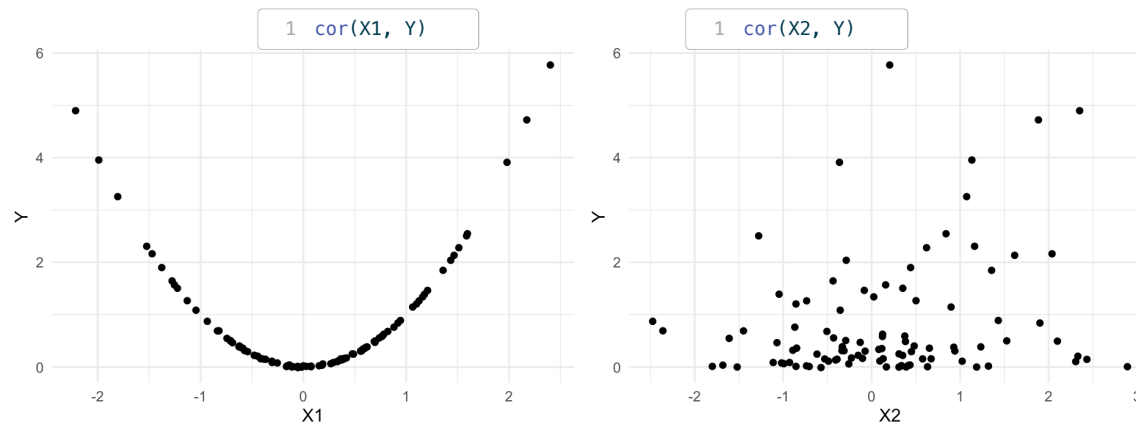

Correlation Cont.

➤ Correlation and Covariance only measure the linear relationship between two variables.

Example

Consider the variables X1 and X2 and their relationship with Y. Which variable has a stronger relationship with Y?

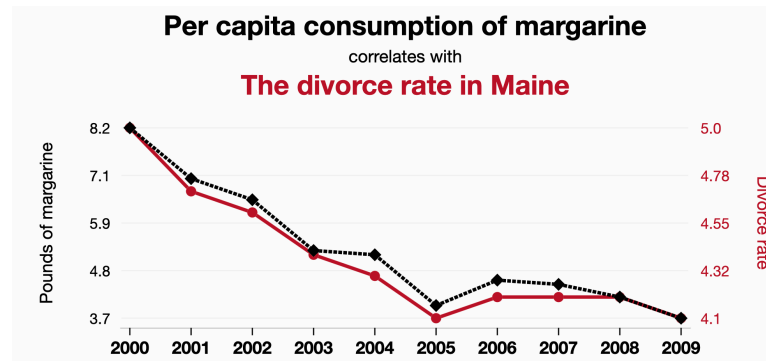
► Show code



However, correlation can only detect linear relationships.

Correlation vs Causation

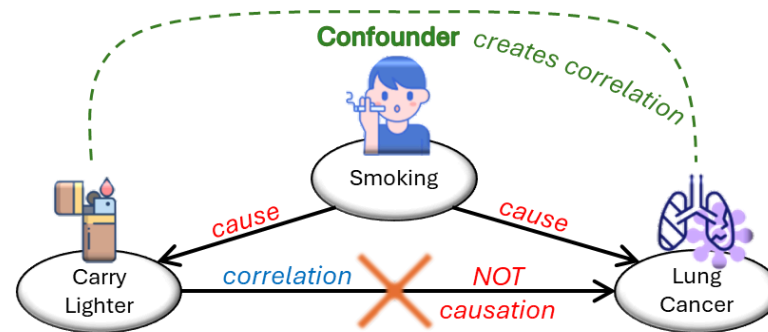
- Correlation only indicates a linear relationship between two variables and does not imply one variable causes the other
- Correlation can imply silly relationships



<https://www.tylervigen.com/spurious-correlations>

Correlation vs Causation Cont.

- Unintentional correlation is often due to a *confounding* variable (Z)
- **Confounding Variable:** a variable that affects both X and Y, making them appear related even if there is no direct connection.



source: Causal analysis overview

- **Causal Inference** refers to determining the causal relationship (if there is any) between X and Y, while properly adjusting for the confounder Z

The End
